

Broadcast Technology UG1

Media Distribution Networks

Media Distribution Platforms

- Media Distribution
 - The TV Headend
 - DVB – Digital Video Broadcasting
 - Transport Stream Multiplex
- Distribution Network Architectures
 - Satellite
 - Digital Terrestrial (DTTV)
 - Cable
- HTTP Streaming: MPEG-DASH
- Higher Definition
- Content Delivery Networks

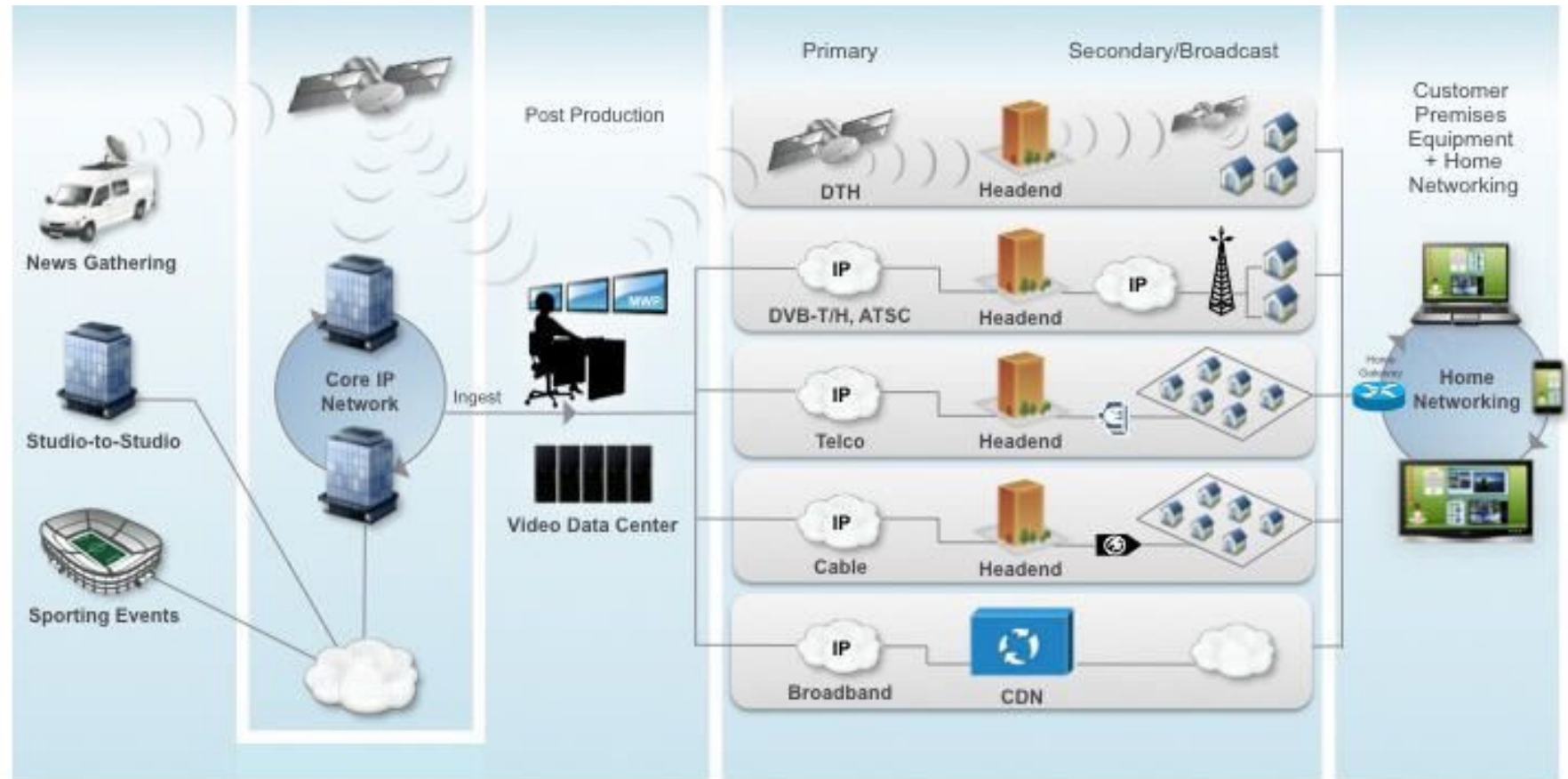
Contribution and Distribution

Production

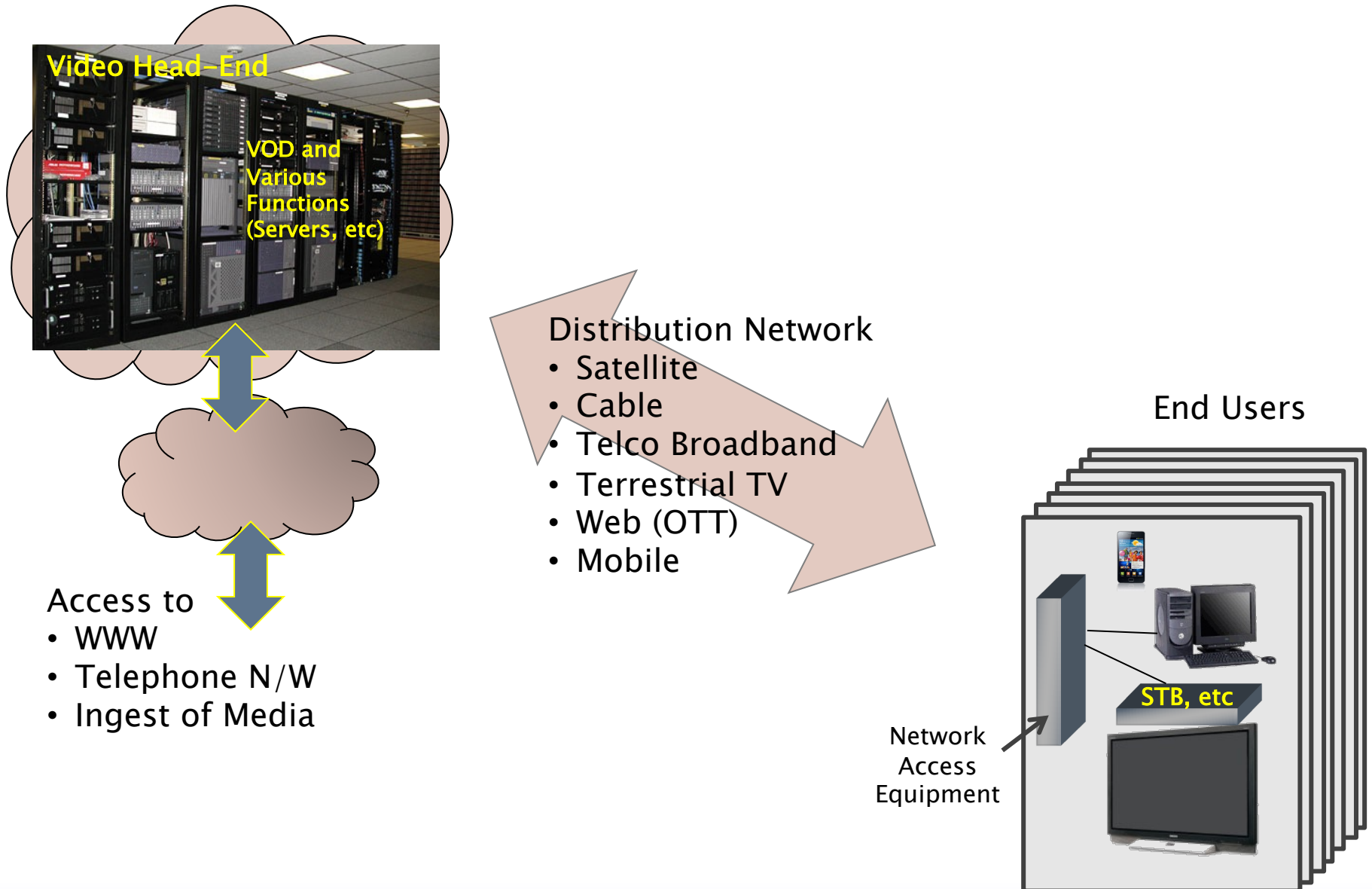
Contribution

Distribution

Consumption

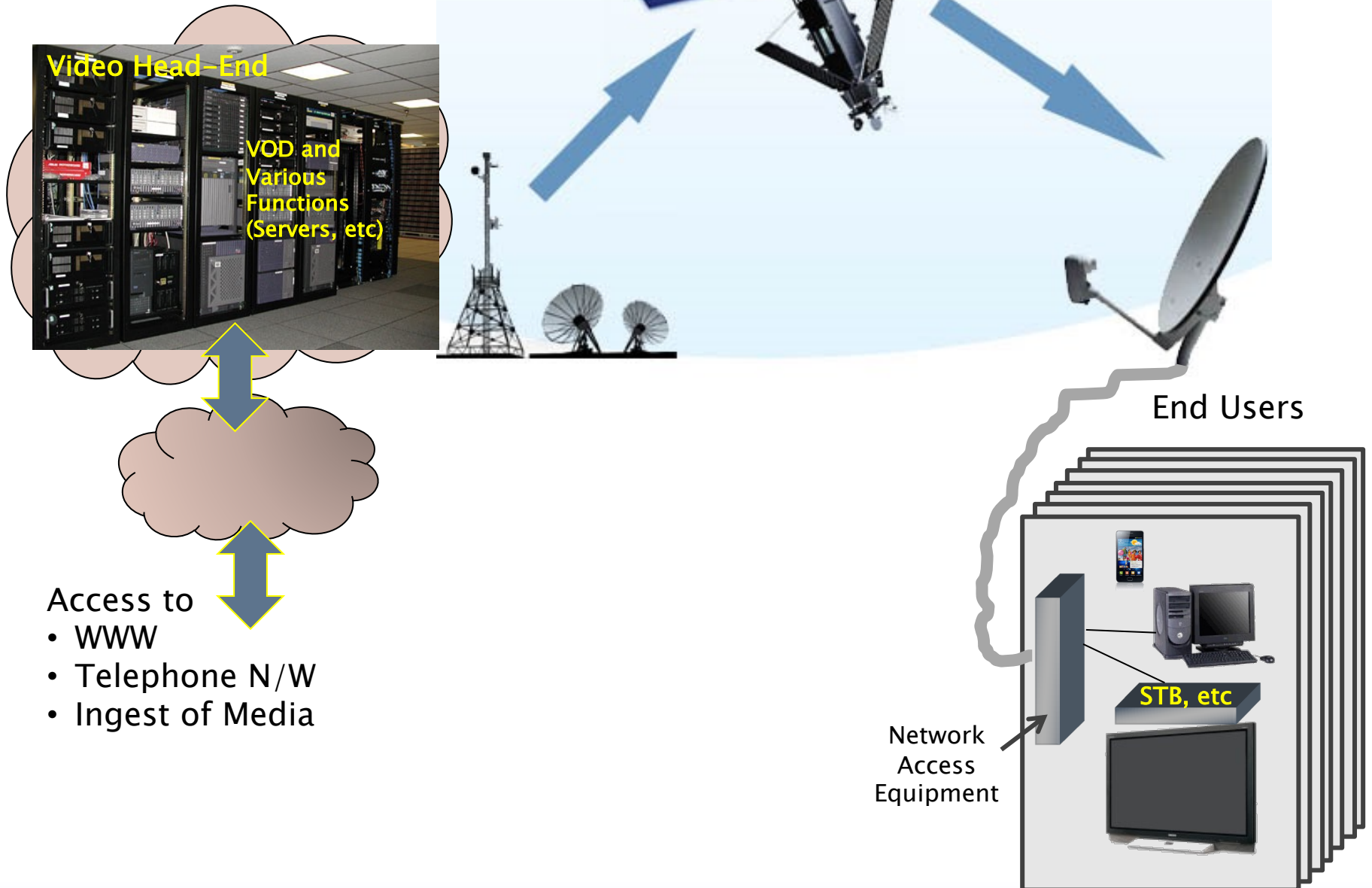


Media Distribution

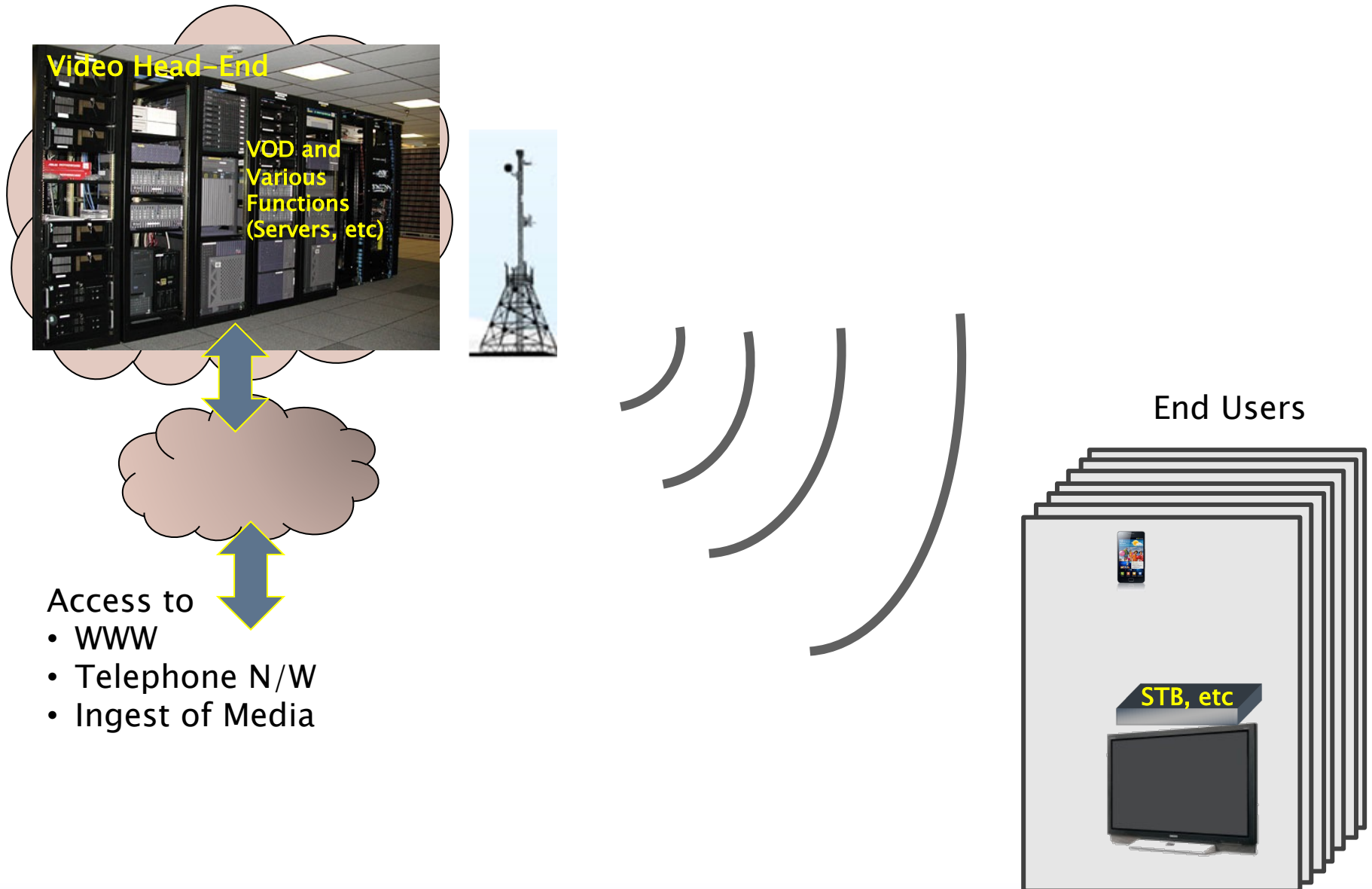


Media Distribution ...

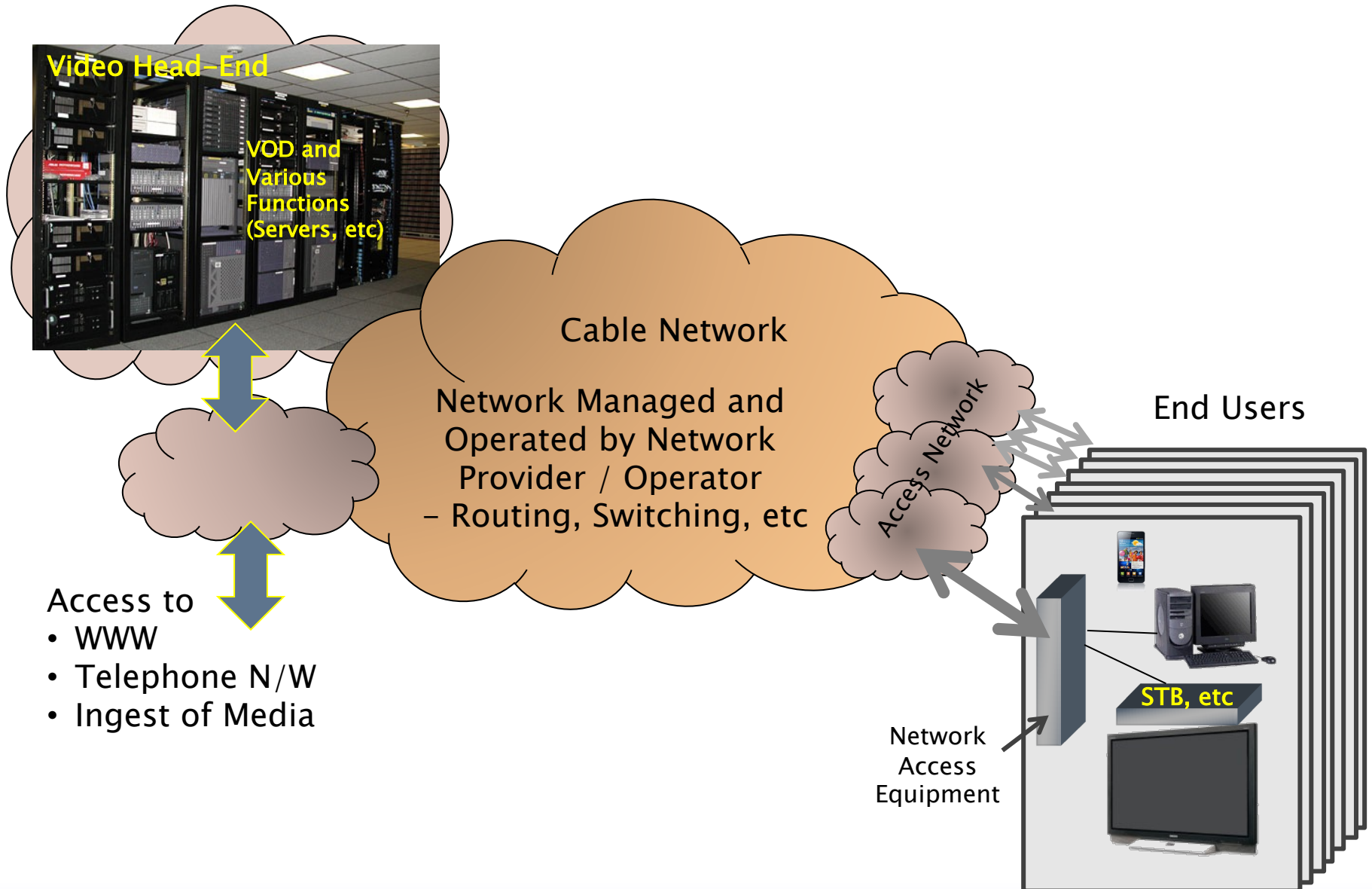
... Satellite



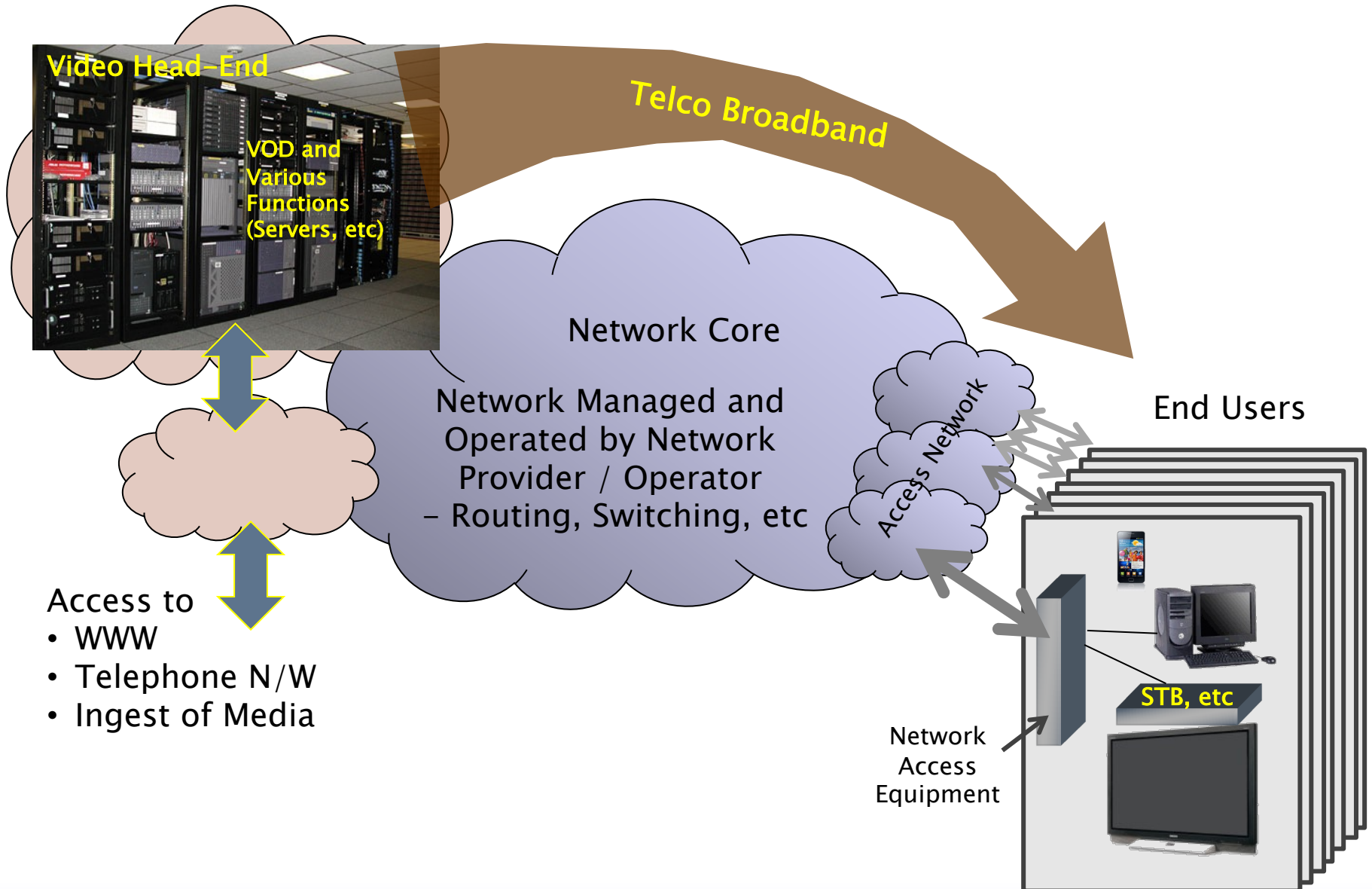
Media Distribution – Digital Terrestrial (DTTV)



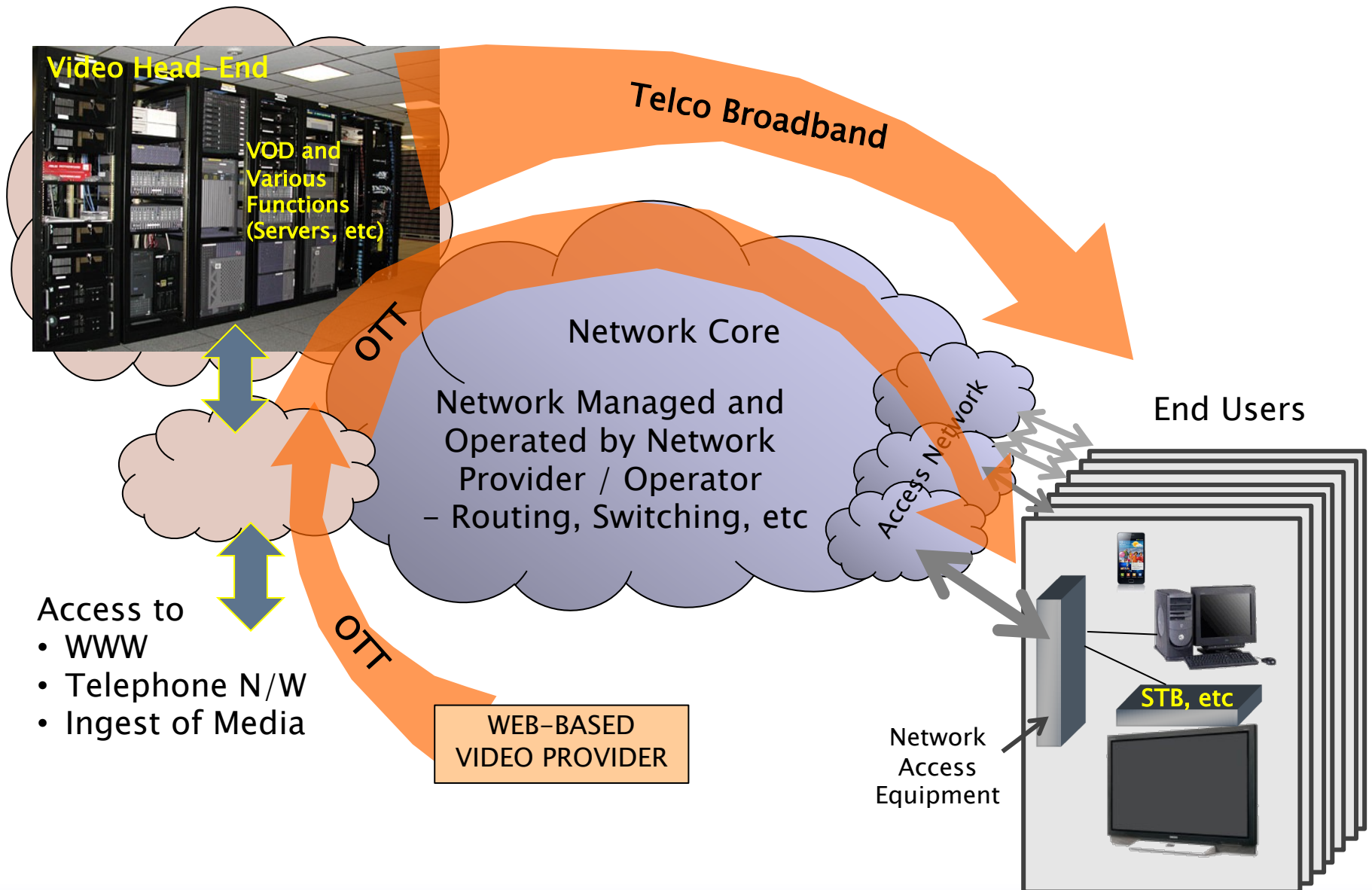
Media Distribution – Cable



Media Distribution – IPTV – Telco Broadband, OTT/Web



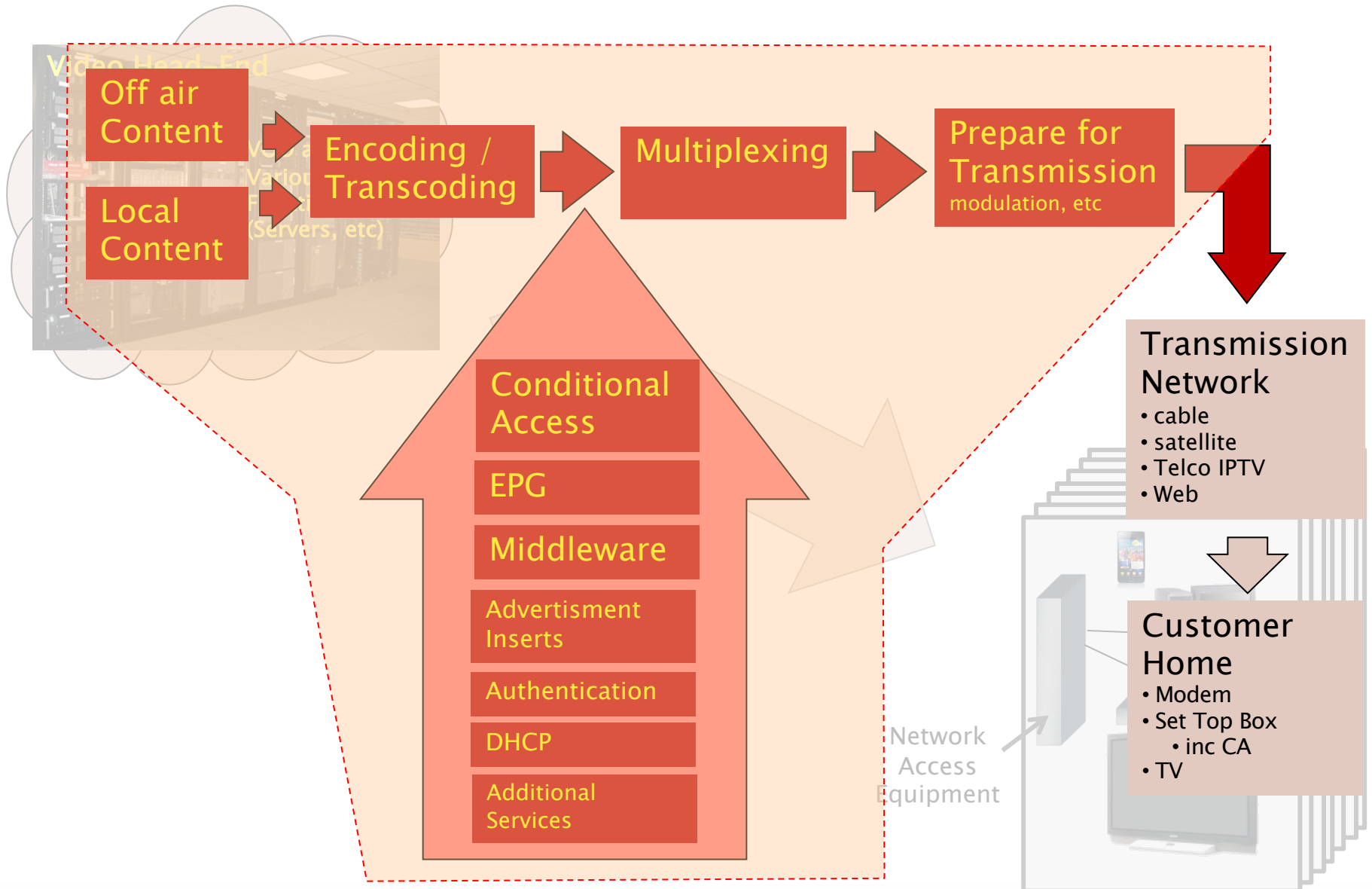
Media Distribution – IPTV – Telco Broadband, OTT/Web



Media Distribution

- Broadcast
 - but what is “broadcast?”
 - ... vs On Demand
 - (what are “video servers”? – or any other server)
 - digital cinema distribution
- Streaming:
 - DVB – Cable, Satellite, DTTV
 - IP-based – IPTV (Telco broadband, OTT)
 - http
 - MPEG → DASH (dynamic adaptive streaming on http)
 - HLS (http live streaming, Apple implemented)
 - rtp protocol

The Headend (Cable, Satellite, Web, Telco ...)



DVB – Digital Video Broadcasting



Industry Organisation

Standardisation Process –

Joint Technical Committee (JTC) of:

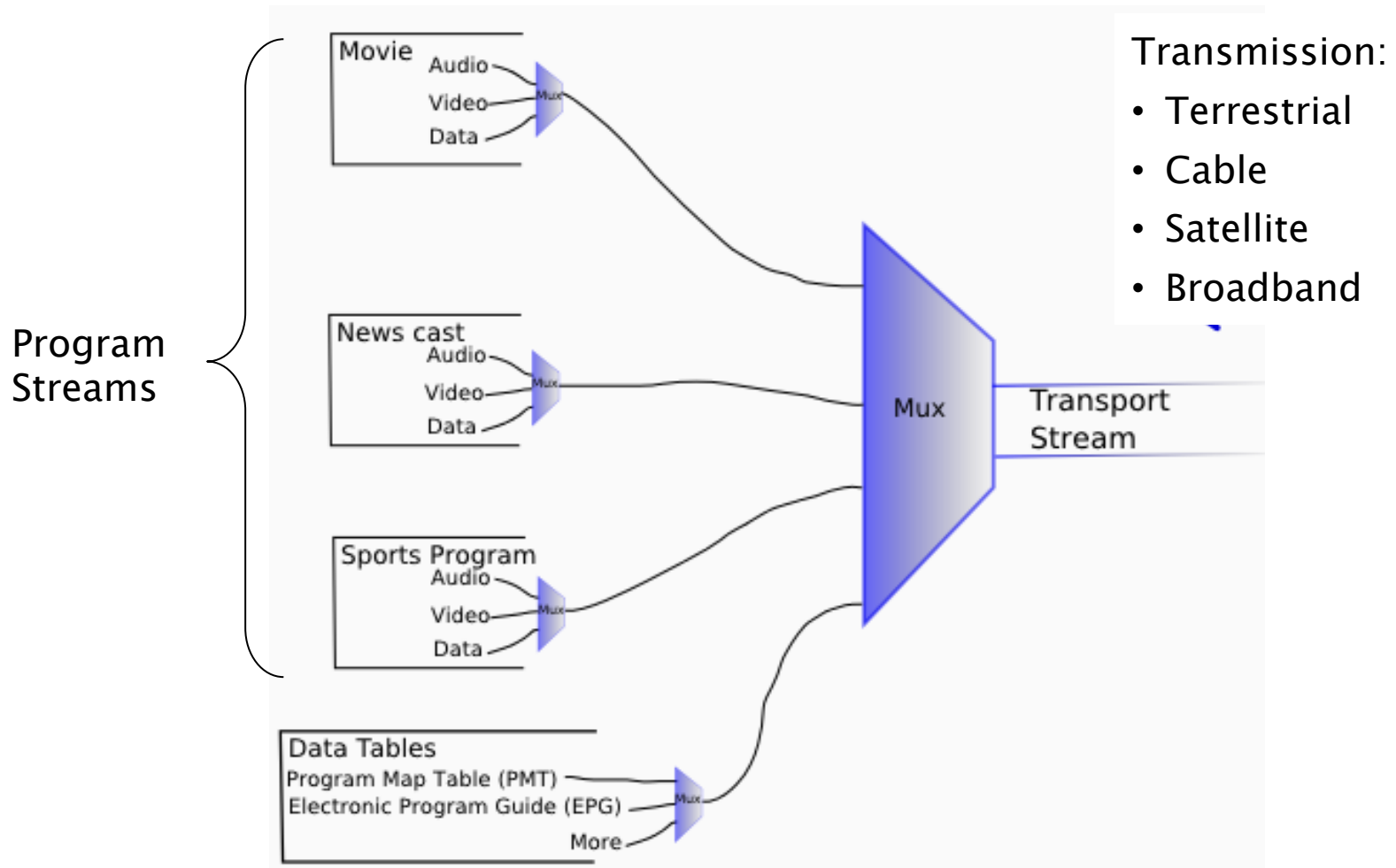
- ETSI – European Telecommunications Standards Institute
 - CENELEC – European Committee for Electrotechnical Standardization CENELEC
 - EBU – European Broadcasting Union
- Note standards refer to other standards and patents, e.g. MPEG

Standards

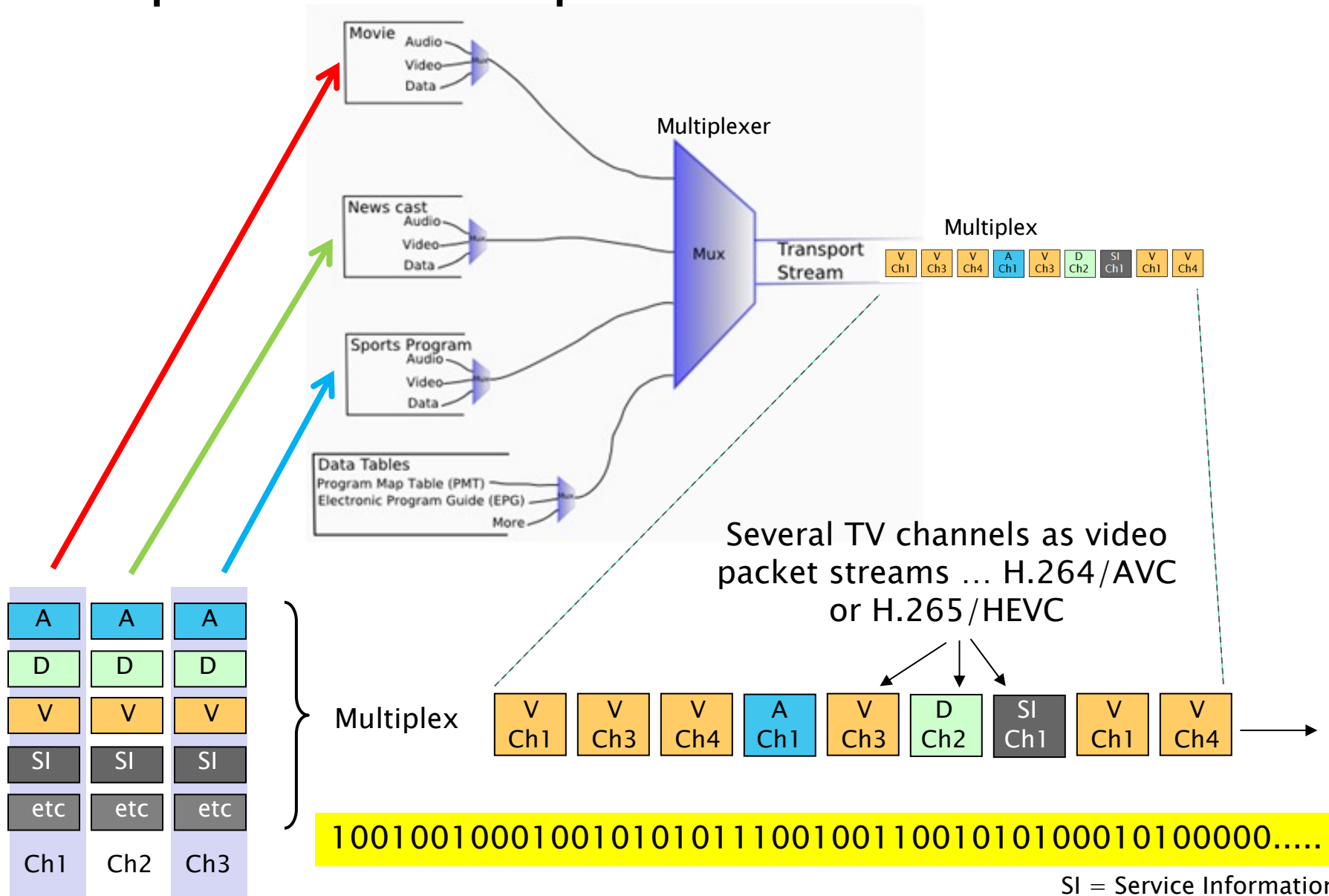
Satellite	DVB-S, DVB-S2, SH
Cable	DVB-C, DVB-C2
DTT	DVB-T, DVB-T2
DTT for handhelds	DVB-H, DVB-H2

DVB-3DTV – Frame Compatible Plano-Stereoscopic 3DTV
DVB-C2 – 2nd Generation Cable Transmission
DVB-CID – Carrier Identification for Satellite Transmission
DVB-CPCM – Content Protection and Copy Management
DVB-FF – DVB File Format
DVB-GEM – Globally Executable MHP
DVB-GSE – Generic Stream Encapsulation
DVB-H – Broadcasting to Handheld Devices
DVB-IPDC – Internet Protocol Datacasting
DVB-IPTV – Internet Protocol Television
DVB-MHP – Open Middleware for Interactive TV
DVB-PCF – Portable Content Format
DVB-RCS2 – Return Channel Satellite
DVB-RCT – Return Channel Terrestrial (03/2001)
DVB-S2 – 2nd Generation Satellite Broadcasting
DVB-SH – Satellite Services to Handheld Devices
DVB-T – Digital Terrestrial Television
DVB-T2 – 2nd Generation Terrestrial Broadcasting
DVB – Emergency Warning System
DVB Project – An Introduction
Hierarchical Modulation in DVB-T (03/2000)
plus many others (inc addenda, etc) ...

Transport Stream Multiplex



Transport Stream Multiplex



Distribution Network Architectures

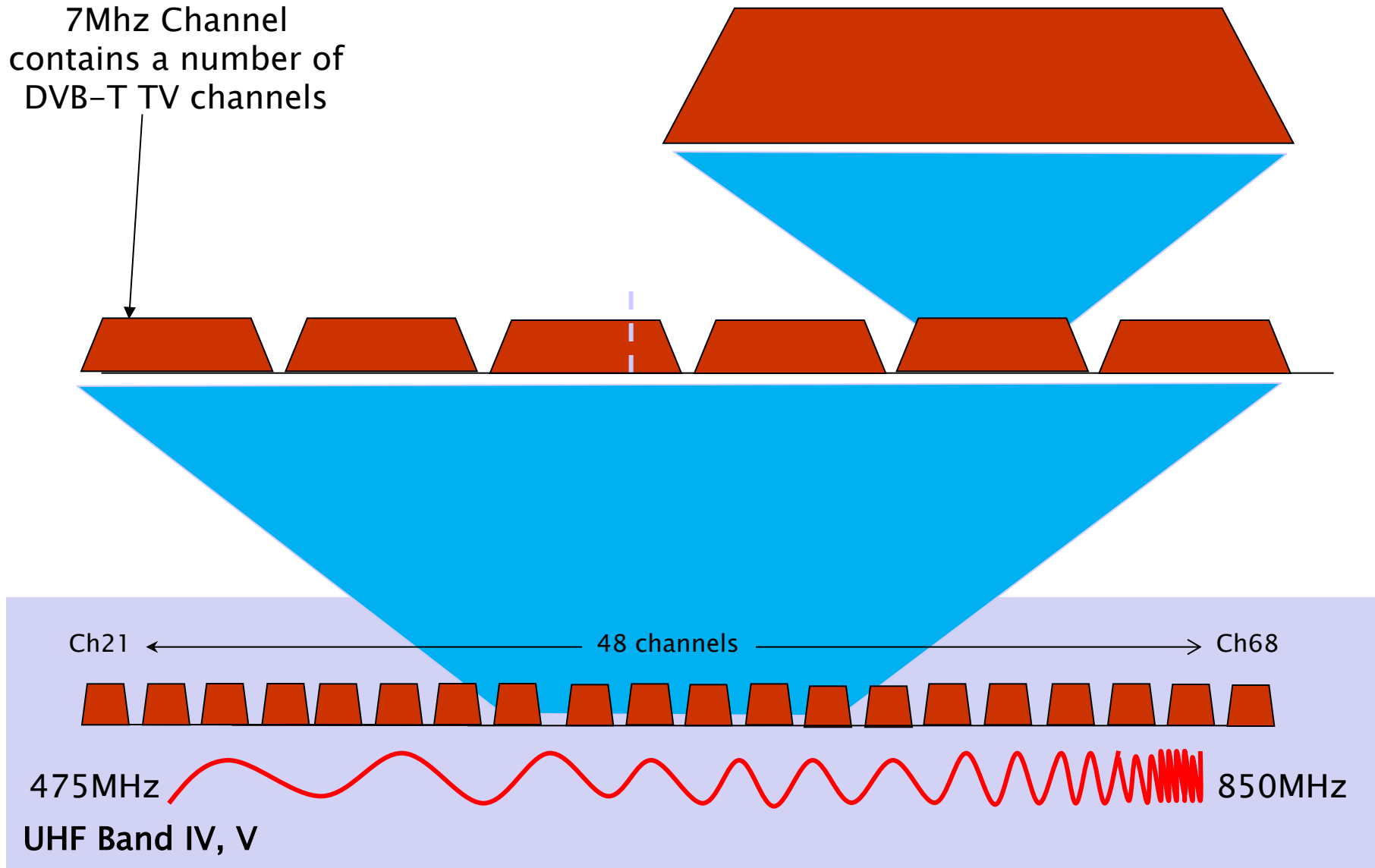


Digital Terrestrial TV (DTTV, e.g. Freeview)



Digital Terrestrial TV (DTTV, e.g. Freeview)

7Mhz Channel
contains a number of
DVB-T TV channels



Freeview – Sutton Coldfield Transmissions (9 Multiplexes altogether)

Mux	Aerial position	Frequency	Height	Mode	Watts
PSB1 BBCA	horizontal max	C43 (650.0MHz)	433m	64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W



1 BBC One West Midlands, 2 BBC Two England, 7 BBC Three, 9 BBC Four, 70 CBBC Channel, 71 CBeebies, 80 BBC News, 81 BBC Parliament, 301 BBC Red Button 1, plus 12 others

PSB2 D3+4	horizontal max	C46 (674.0MHz)	433m	64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
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3 ITV (Central (West micro region)), 4 Channel 4 Midlands ads, 5 Channel 5 Part Network ads, 6 ITV 2, 13 Channel 4+1 Midlands ads, 14 More 4, 15 Film 4, 28 E4, 33 ITV +1 (Central west),

PSB3 BBCB	horizontal max	C40+ (626.2MHz)	433m	256QAM 32KE 2/3 40.2Mb/s DVB-T2 MPEG4	200,000W
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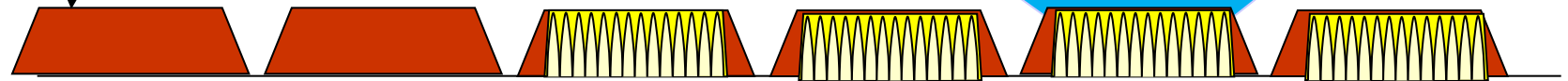
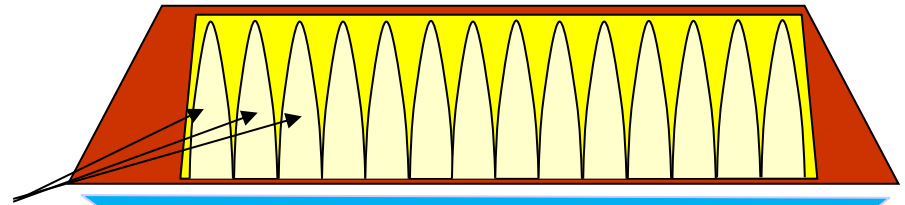
73 CBBC HD, 101 BBC One HD (England no regional news), 102 BBC Two HD (England), 103 ITV HD (ITV Central West), 104 Channel 4 HD Midlands ads, 105 BBC Three HD, 303 BBC Red Button HD, plus 1 others

<http://www.ukfree.tv/txdetail.php?a=SK113003>

Digital Terrestrial TV (DTTV, e.g. Freeview)

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DVB-T TV channels

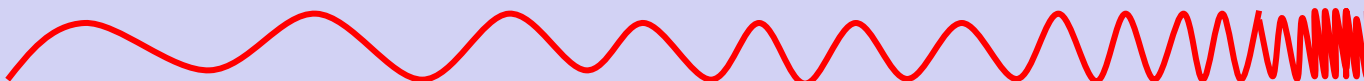
- a transport stream
modulated on many carriers



Ch21 ← 48 channels → Ch68



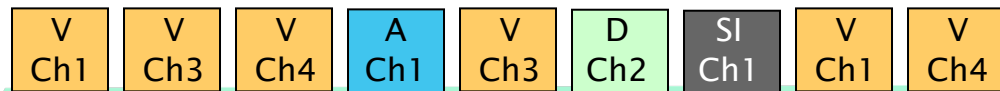
475MHz



850MHz

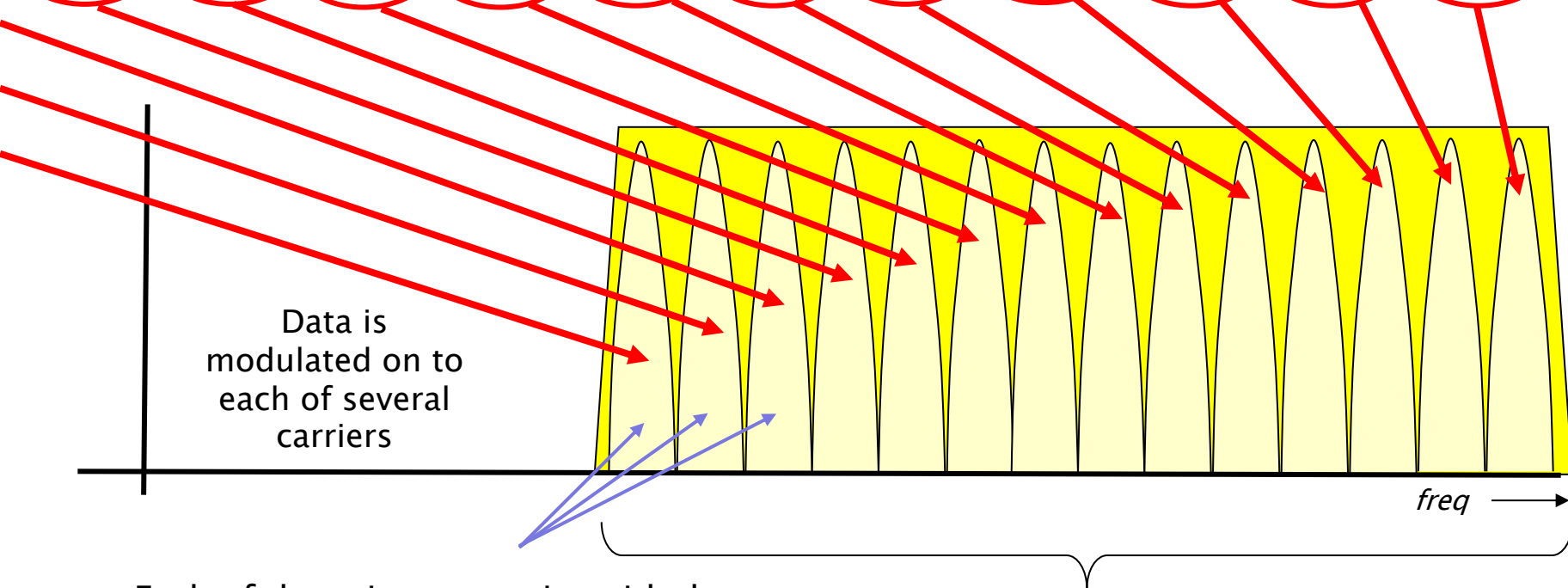
UHF Band IV, V

Multiplexing the Channels



Total data to be transmitted

00100010010101011100100100010010101110010011001010100010100000.....



Data is
modulated on to
each of several
carriers

Each of these is one carrier with data
modulated on to it in *Quadrature
Amplitude Modulation*
(QAM – a mix of amplitude and phase
modulation)

All these carriers are separate signals and
are all modulated on to one main carrier

Satellite Distribution

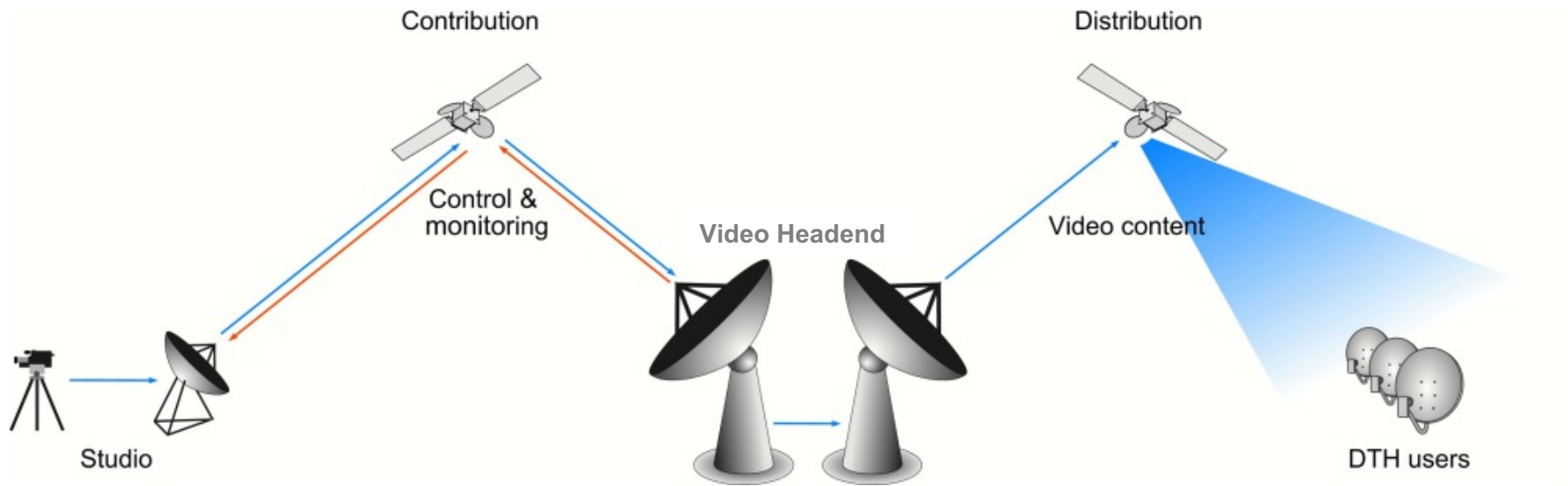


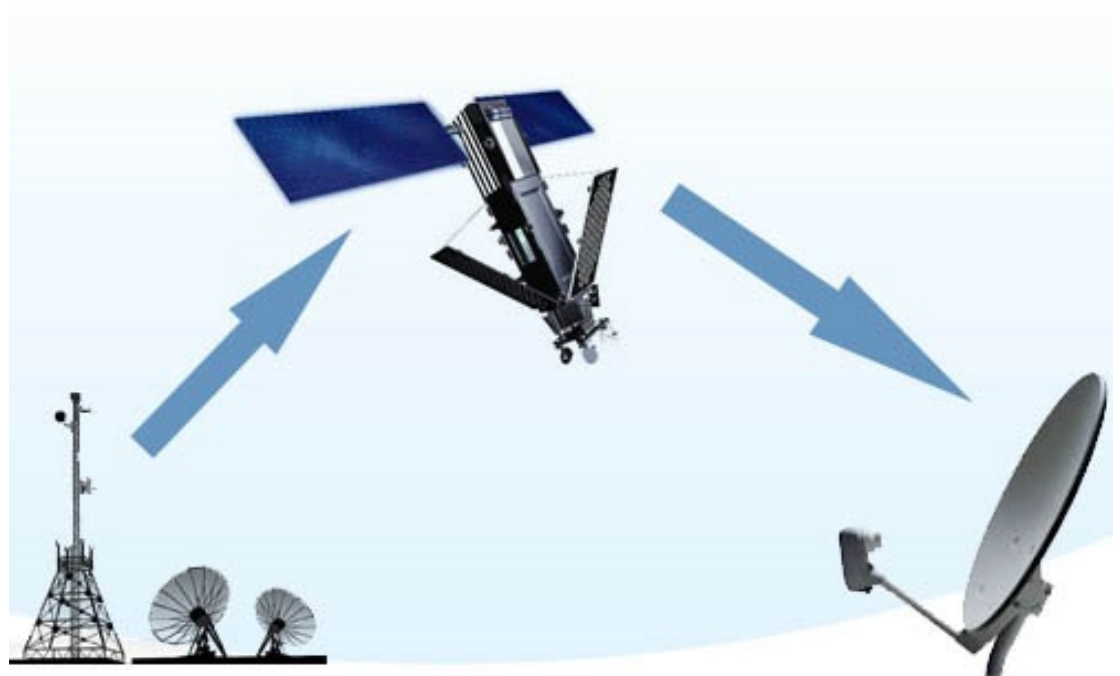
image from
http://upload.wikimedia.org/wikipedia/en/8/80/Sat_contribution_fig.jpg

Satellite TV

DBS – Direct Broadcast Satellite; Broadcasting Satellite Service

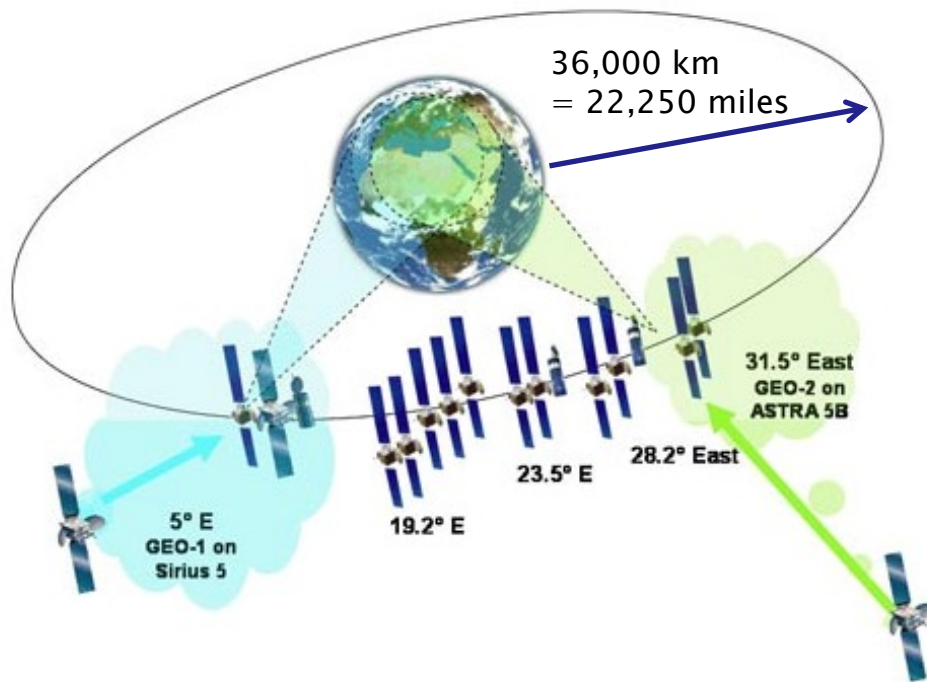
- One-way broadcast
- From a communications satellite to an end user
- Uplink: large dish 9–12m dia
- Receiver – Small round or oval dish antenna < 20" diameter.
- Compressed digital signal in the K_u band
(11–15GHz)

- Reception affected by
 - rain
 - solar activity



<http://www.supersatelliteselection.com/how-satellite-tv-works.html>

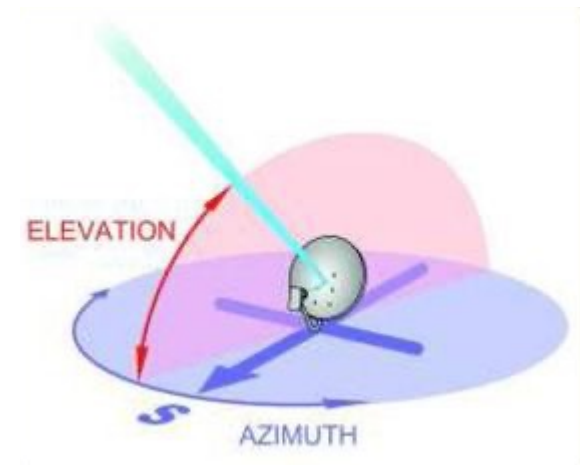
Geostationary Orbit



- appears as a fixed position to an earth-based observer
- revolves around the earth at a constant speed once per day over the equator
- ground based antennae – no need for expensive tracking equipment

Some satellite data ...

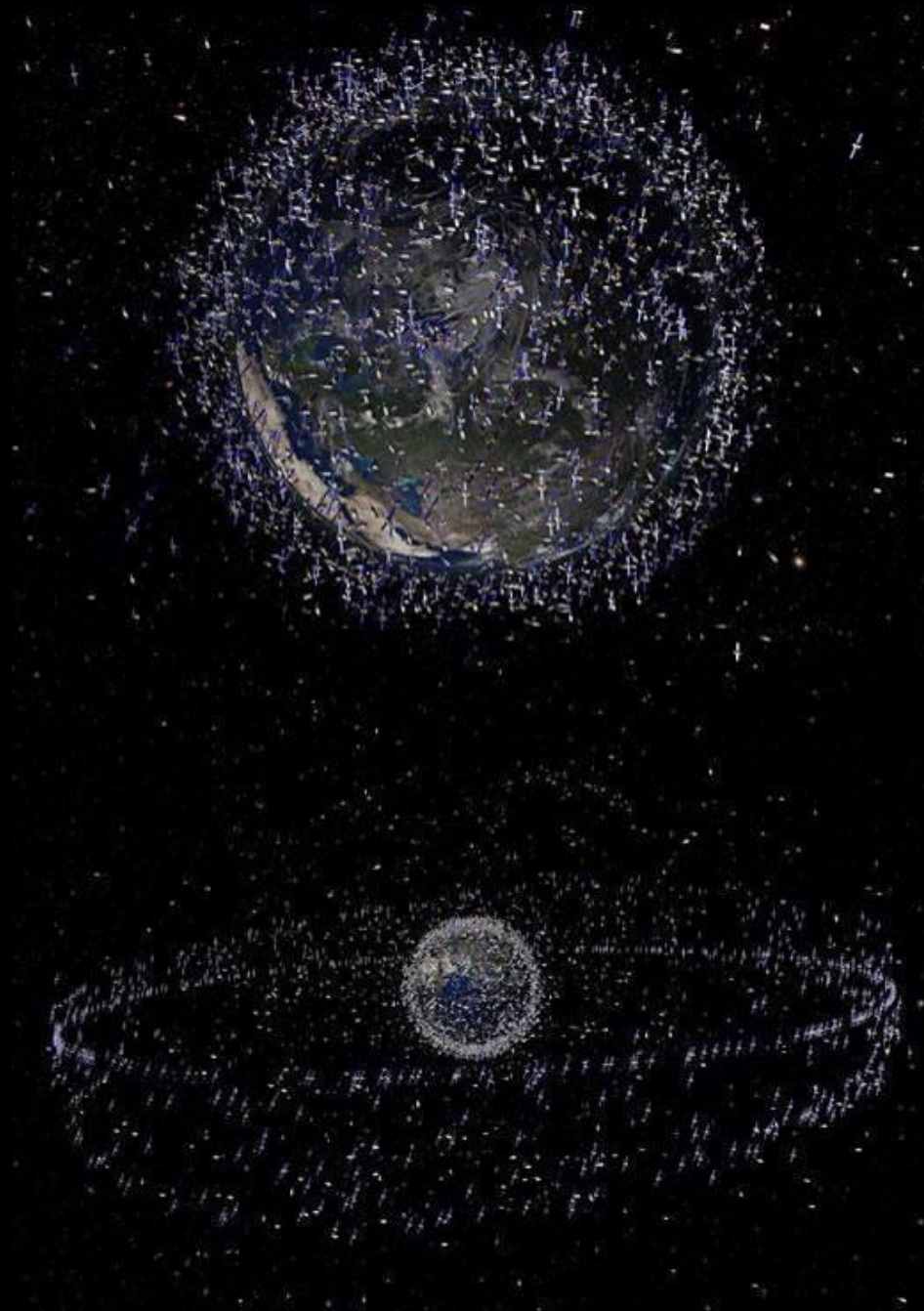
- SES Astra satellites located at 28.2° east (Astra 2A/2B/2C/2D)
- Eutelsat's Eurobird 1 satellite at 28.5°E



receiver dish parameters

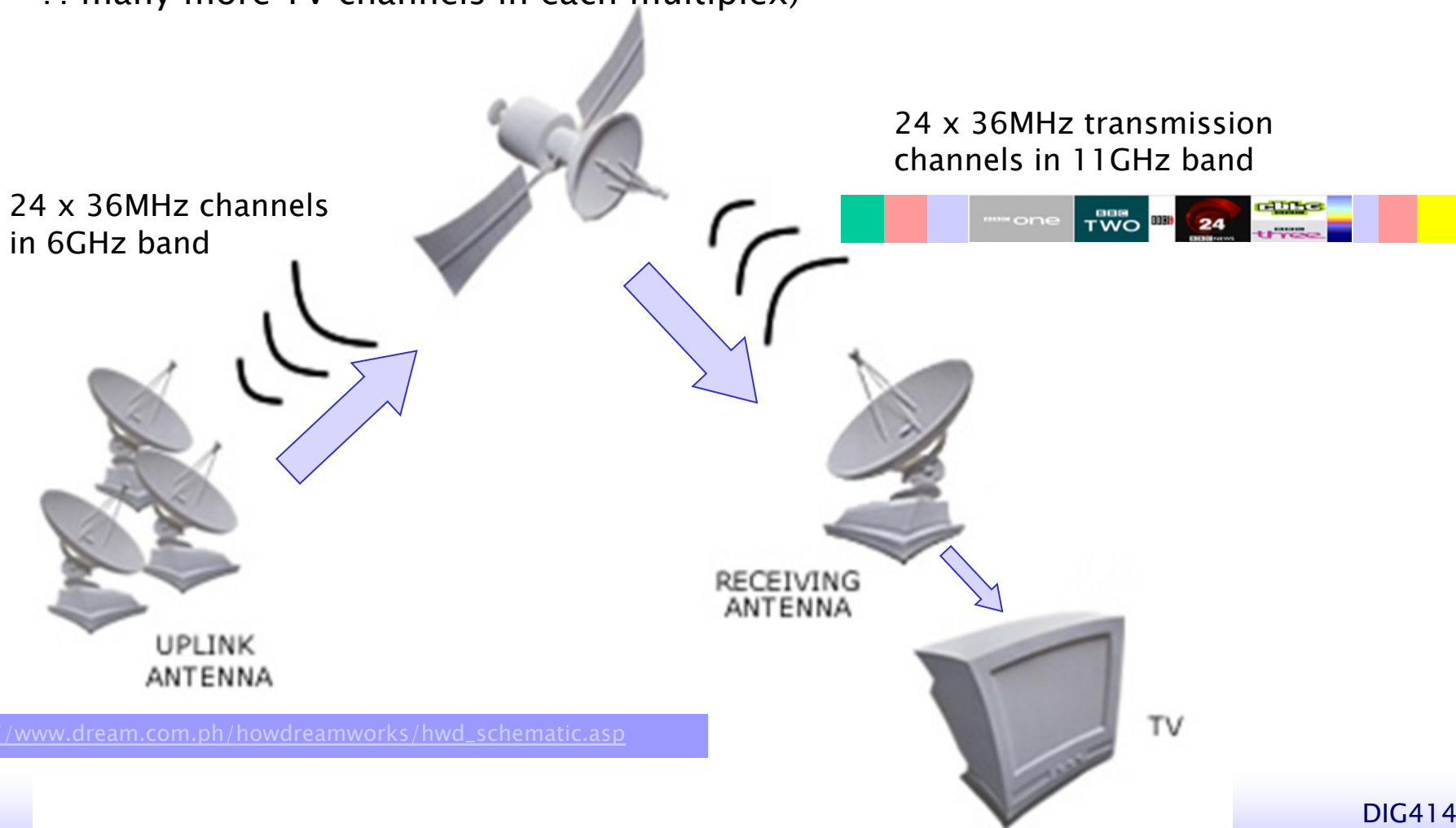
EVERY SINGLE SATELLITE ORBITING THE EARTH

March 2014



Satellite Transponders

- Each Transponder is a separate transceiver or repeater
- Each transmission channel is a multiplex of several digital TV channels
- (note – bigger transmission channels than DTT)
- $\therefore$ many more TV channels in each multiplex)



Cable TV



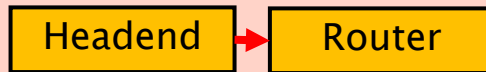
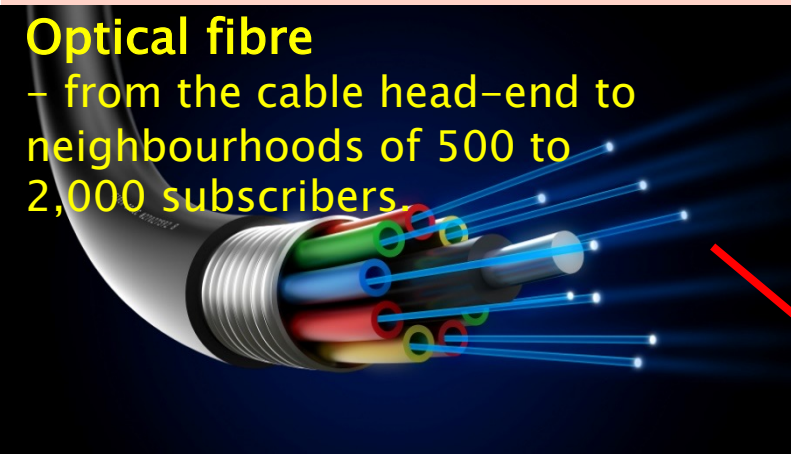
Cable TV

HFC network – hybrid fibre-coax network

Combine both optical-fibre and coaxial cable lines.

Optical fibre

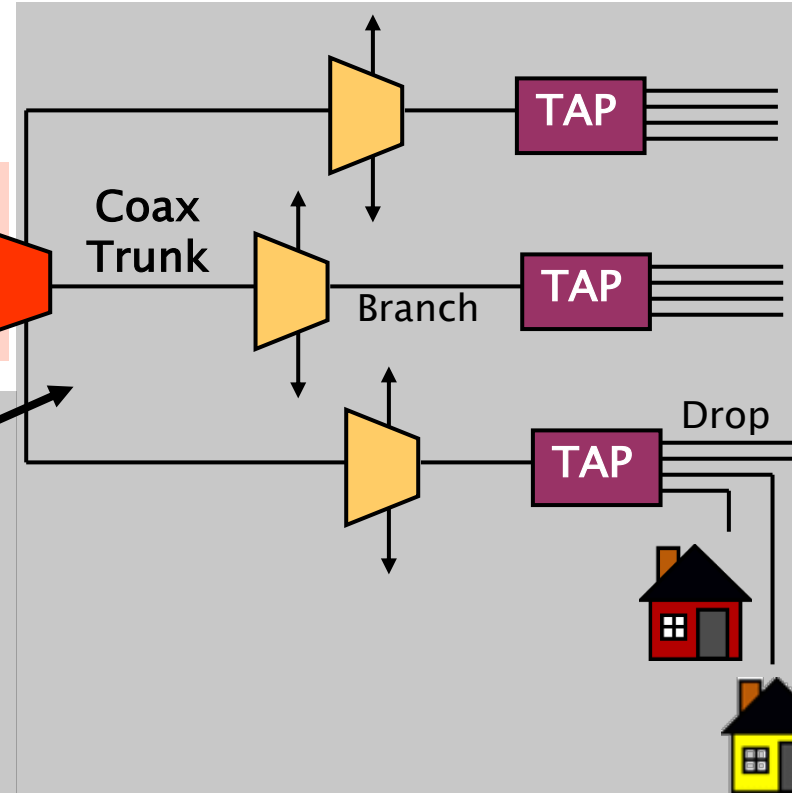
– from the cable head-end to neighbourhoods of 500 to 2,000 subscribers



Fibre Trunk

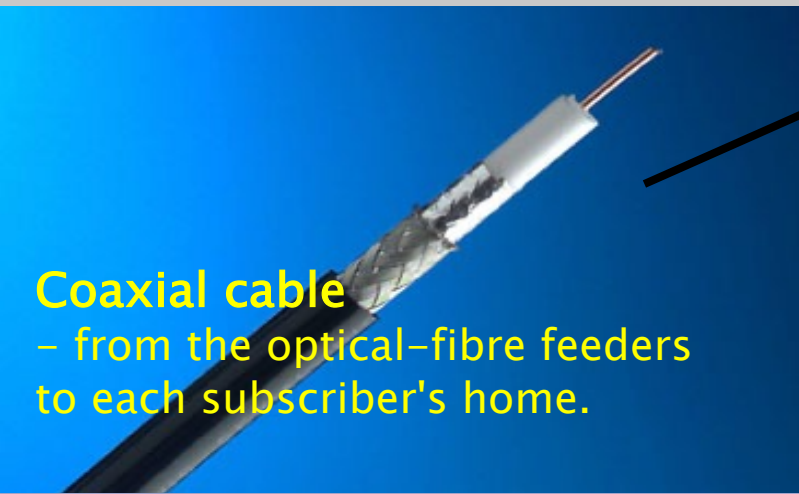
Hybrid networks:

- fibre's reliability and bandwidth benefits
- lower cost than a pure fibre network
- utilising existing copper plant

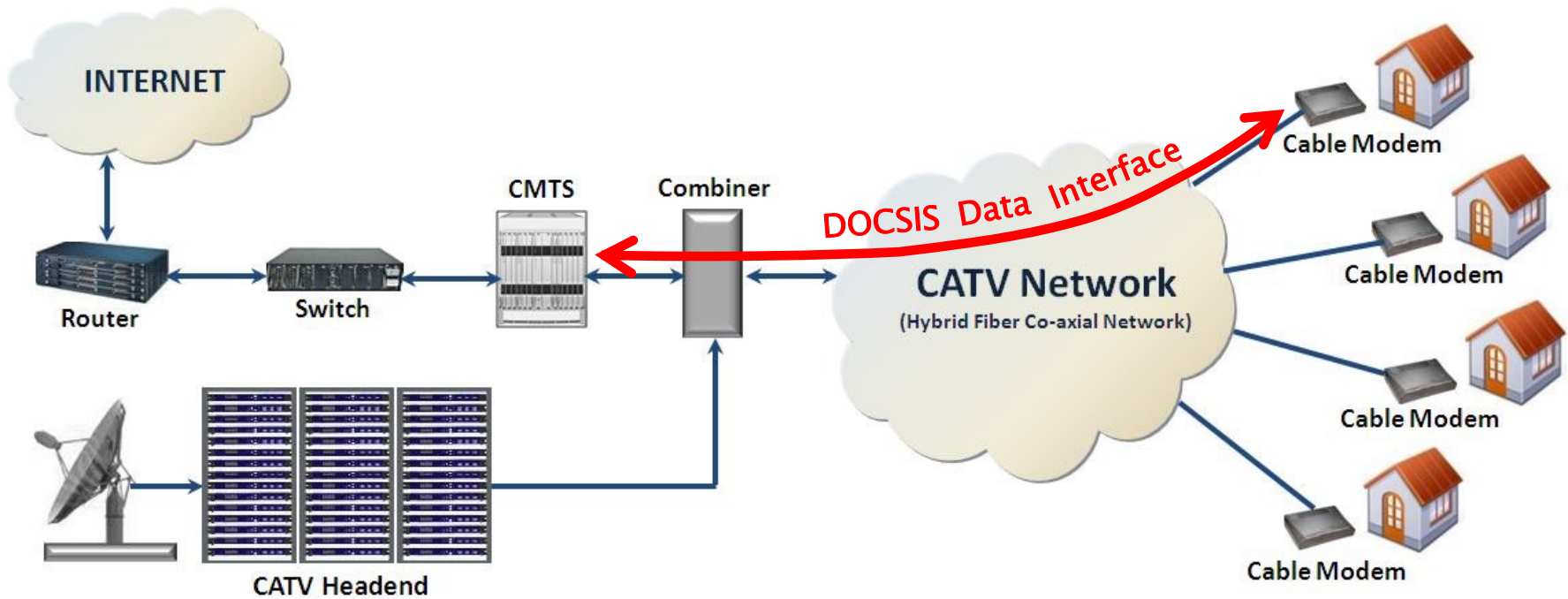


Coaxial cable

– from the optical-fibre feeders to each subscriber's home.



Cable TV – Data and TV to the user



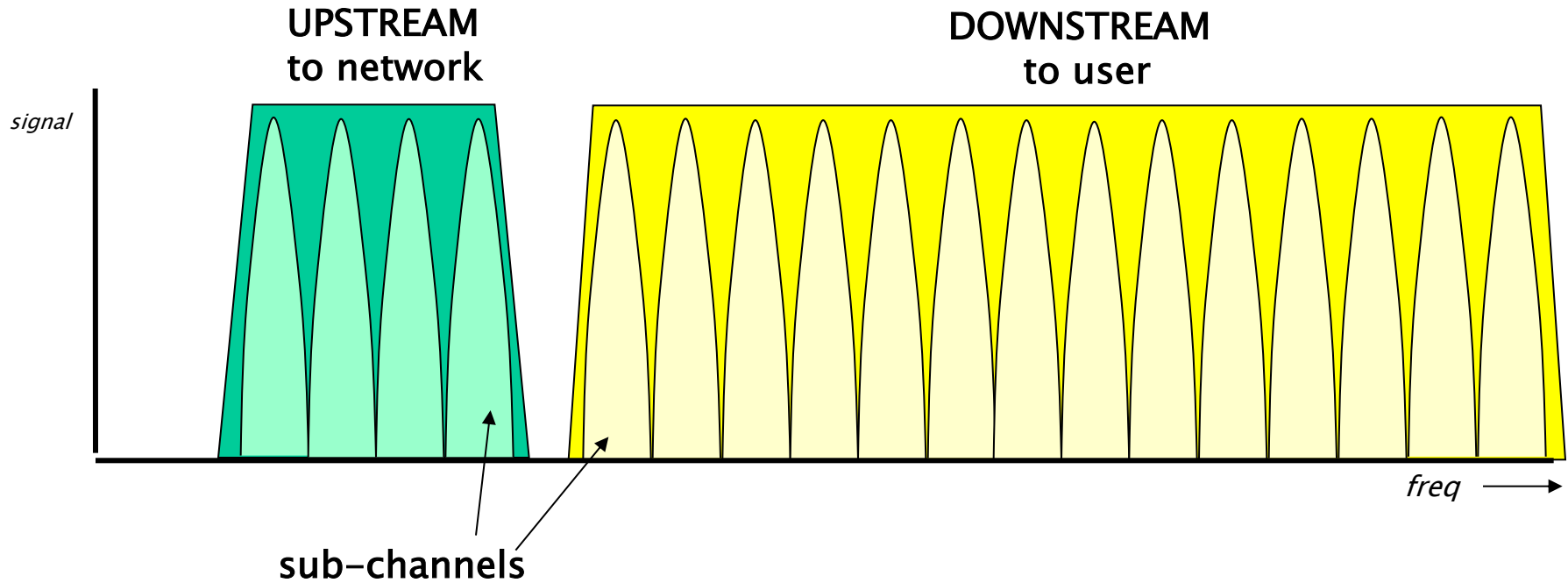
Specifies modulation schemes and protocol

- for exchanging signals over cable
- to carry MPEG and IP

DOCSIS – Data Over Cable Service Interface Specification

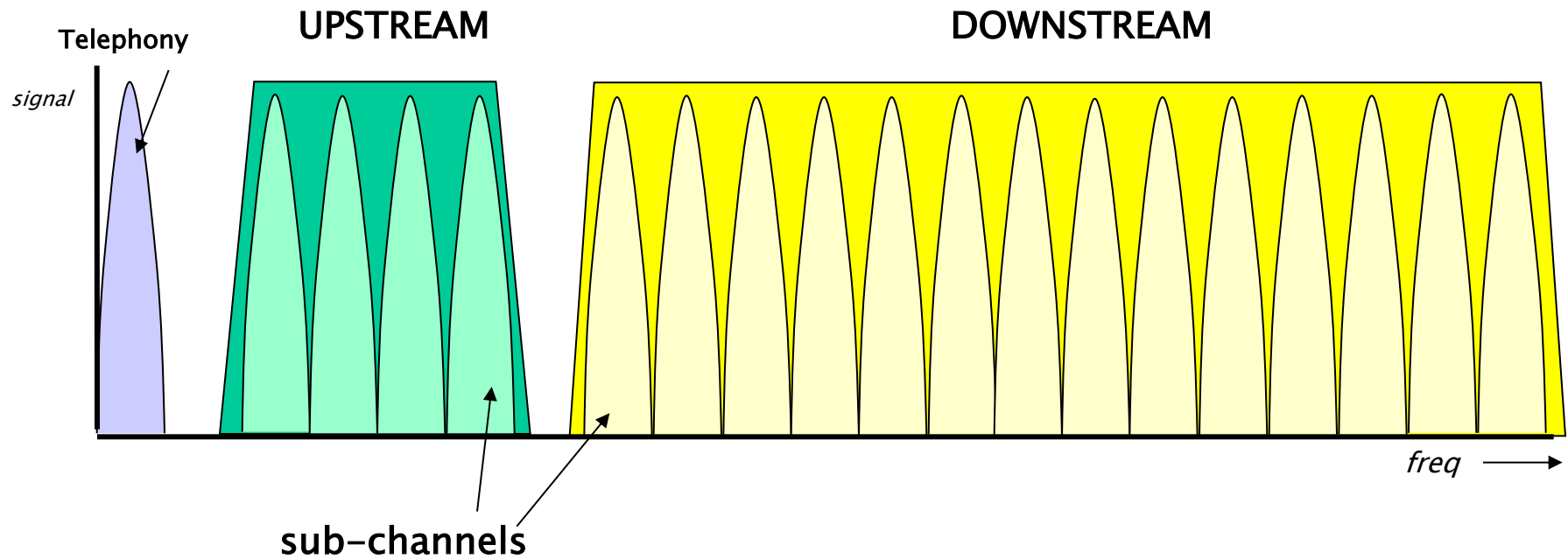
- The Standard for the interface between the CMTS and the Cable Modem
- Developed and licensed by CableLabs

Cable TV – Data and TV to the user



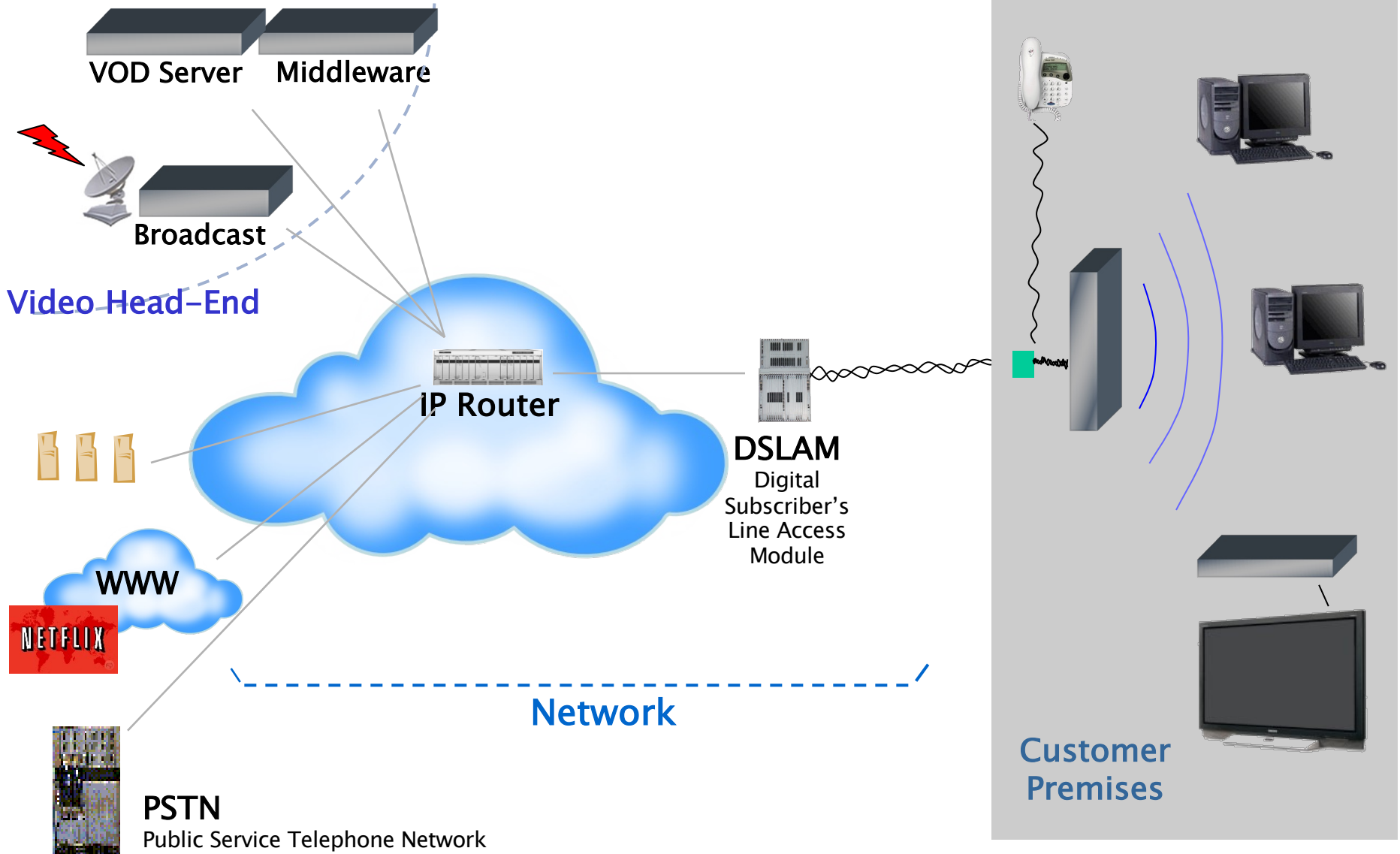
Telecomms Service Providers (Telco) Broadband Access

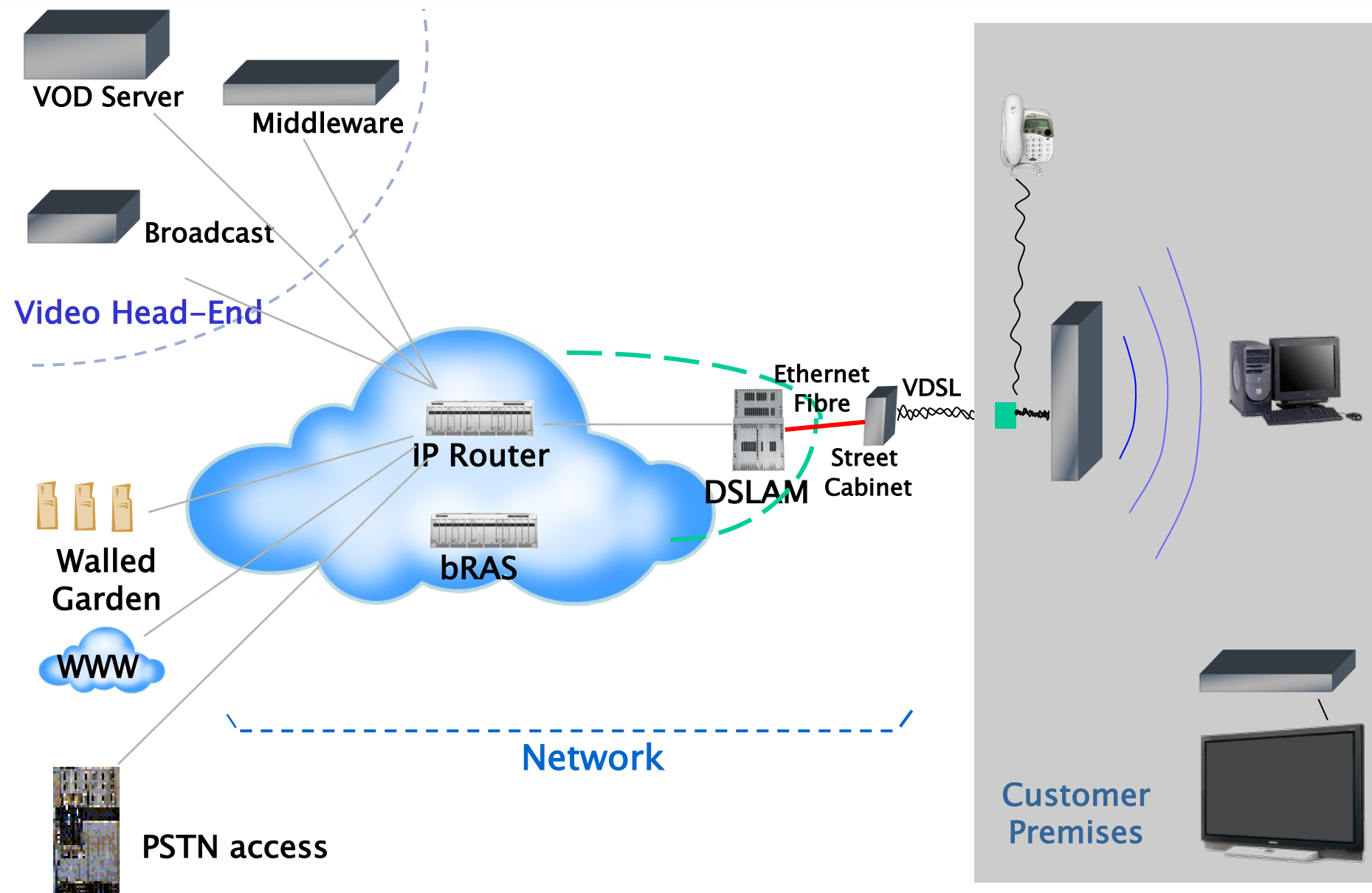
xDSL (ADSL, XDSL)



BT Sport

Telco Broadband (IPTV)

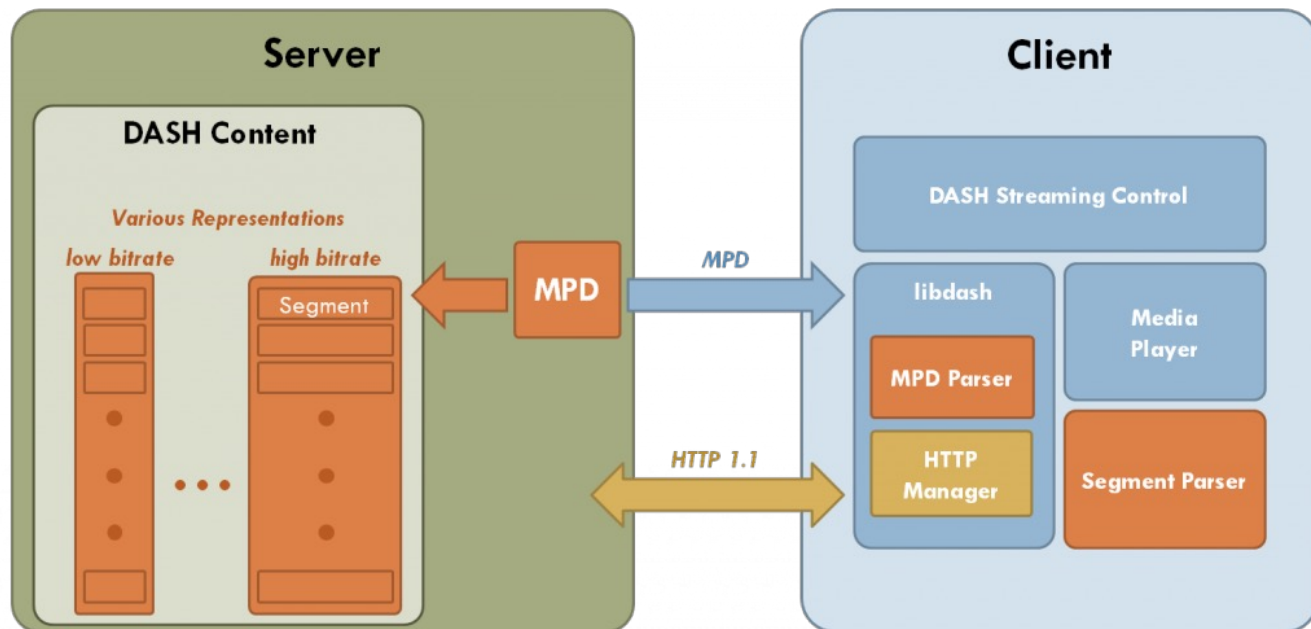




HTTP Streaming: MPEG-DASH

Streaming content from HTTP server to a DASH client

- content stored on HTTP server; delivered using HTTP



Content exists on the server in two parts:

- Media Presentation Description (MPD)
- content presentation types, its various alternatives, their URL addresses, etc
- **Segments** – contain the content bitstreams
 - in the form of chunks, in single or multiple files.

How Much Access Bandwidth? (*up to recently ...*)

Video

(biggest payload)

assume MPEG4 / WM9, say

2x SD streams = 2 x [0.5 ... 1 Mbs]

+ 2 x HDTV streams = 2 x [3 ... 6 Mbs]

= ~ 10 ... 15 Mbs

*Contentious figures!
But quite likely*

Data

assume RT (or better) download video ... ~ 2 ... 3 Mbs

Voice

(negligible)

~ ...

Total

... .. < 20 Mbs

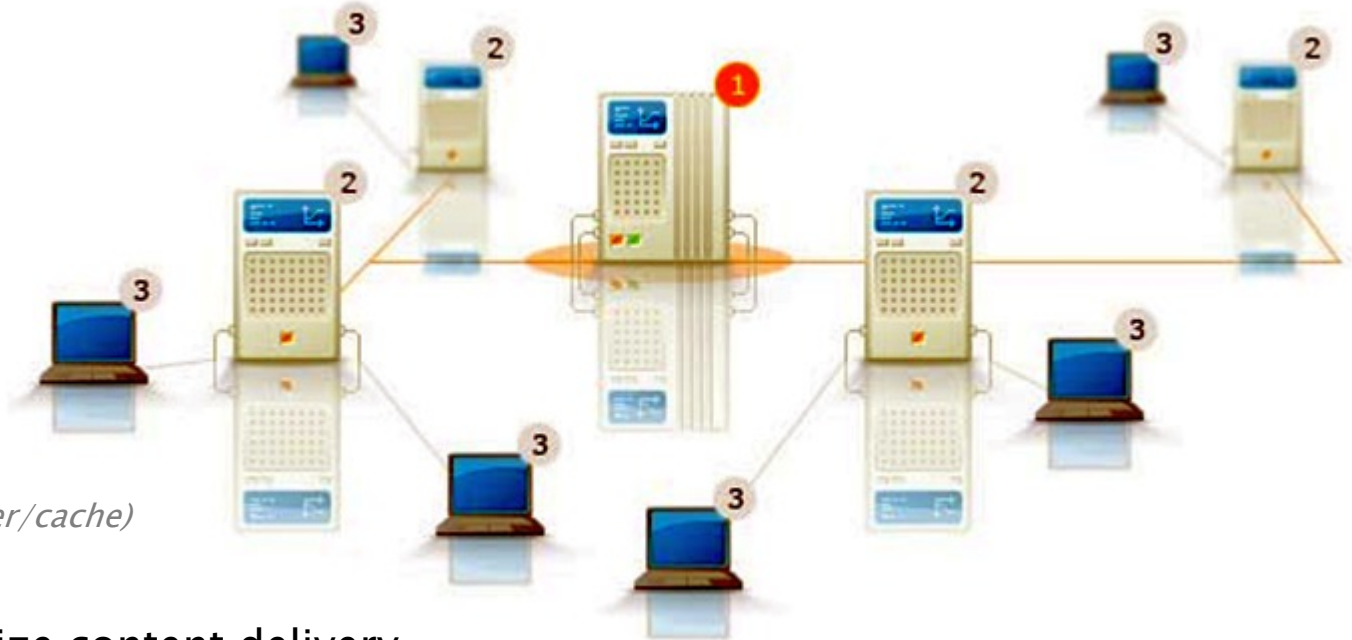
however – UHD, HDR, Frame Rates...

Content Delivery Networks

Content Networking

Internet – end-to-end

- core network is simple – only forwards data packets
- intelligence is at edges



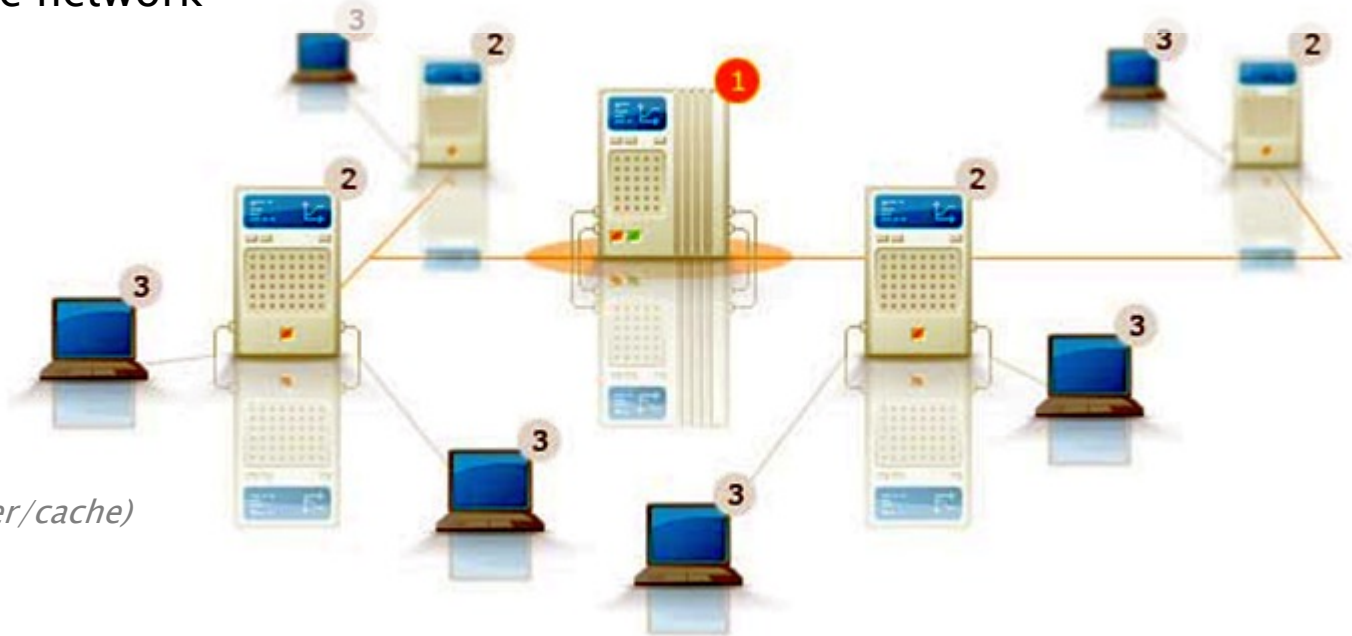
1 – central server
2 – cache (edge server/cache)
3 – end users

CDN – to optimize content delivery

- infrastructure solutions to network service providers who deploy their own dedicated digital media delivery
- asset tracking
- network performance
 - real-time load
 - statistics reporting

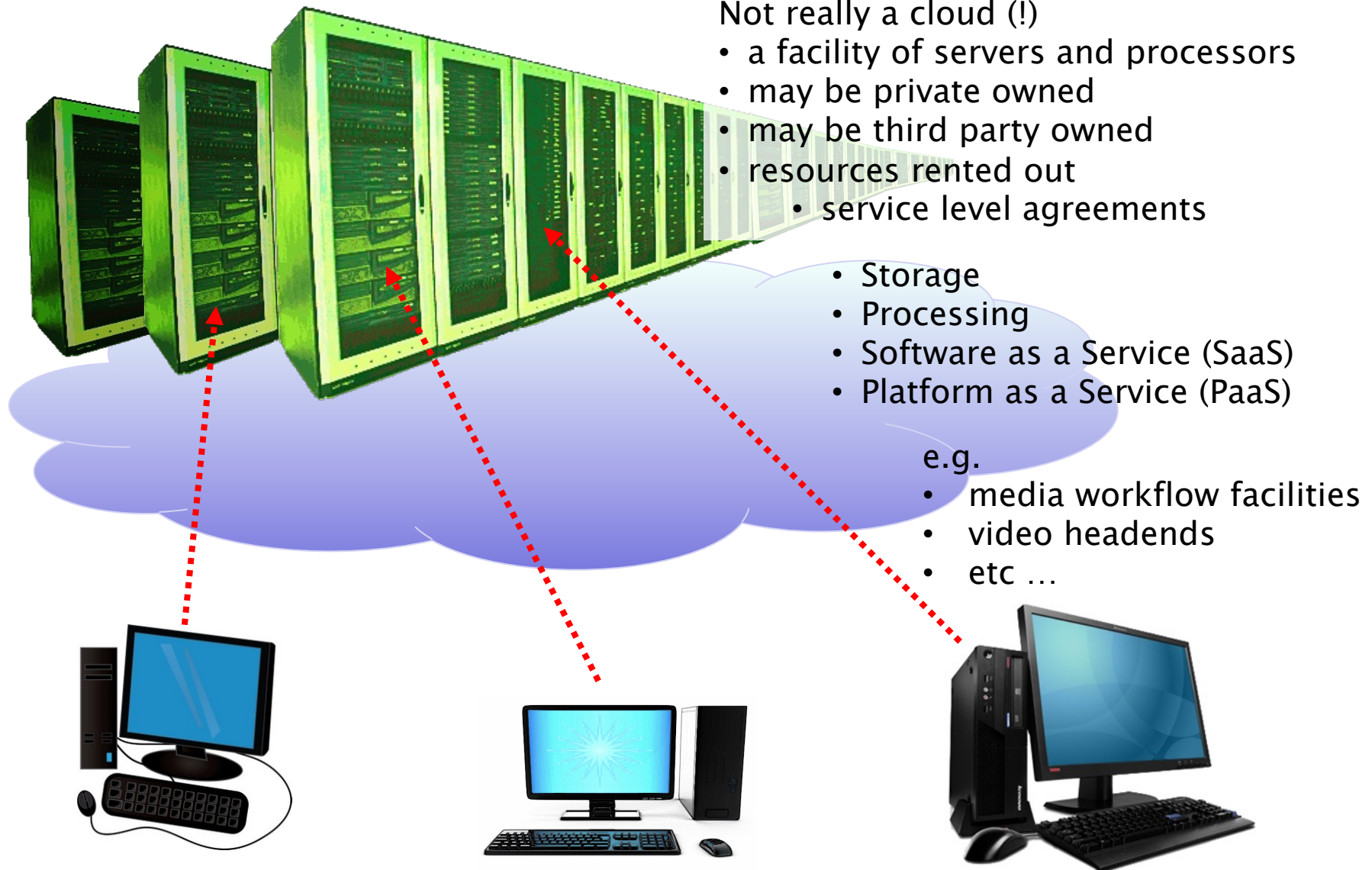
Content Networking

- system of caches containing copies of data
- data is distributed at various points in a network
- maximizes the bandwidth required to access the data from clients throughout the network



*1 – central server
2 – cache (edge server/cache)
3 – end users*

The Cloud



Cloud Computing, Amazon Instances

Instance	EC2 Compute Units	Memory (GB)	Instance Storage (GB)	Platform	I/O Performance	Hourly Price
Small	1 (1 Virtual Core x 1 Compute Unit)	1.7	160	32-bit	Moderate	\$0.10
Large	4 (2 Virtual Core x 2 Compute Unit)	7.5	850	64-bit	High	\$0.40
X-Large	8 (4 Virtual Core x 2 Compute Unit)	15	1690	64-bit	High	\$0.80
High-CPU Medium	5 (2 Virtual Core x 2.5 Compute Unit)	1.7	350	32-bit	Moderate	\$0.20
High-CPU XL	20 (8 Virtual Core x 2.5 Compute Unit)	7	1690	64-bit	High	\$0.80

(more information at – <http://aws.amazon.com/ec2/instance-types>)